ETH zürich

Mobile Computing

Emerging Technologies
IEEE 802.11 Wi-Fi - Physical Layer

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Course Outline

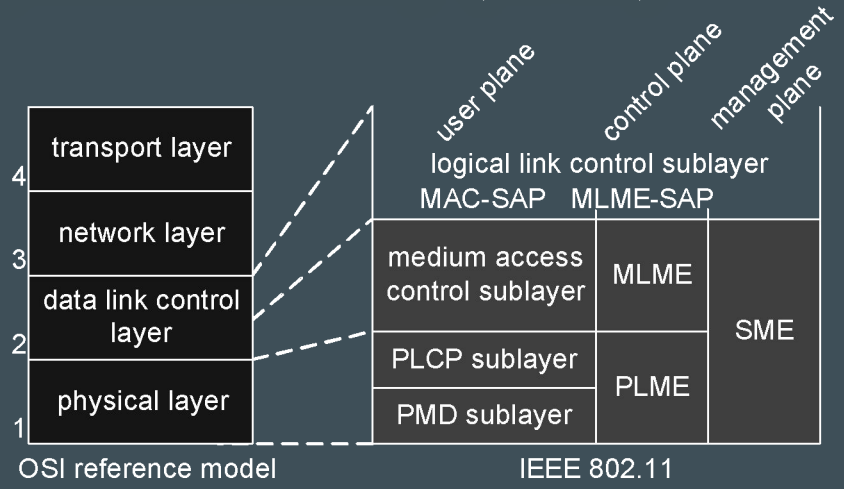
Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

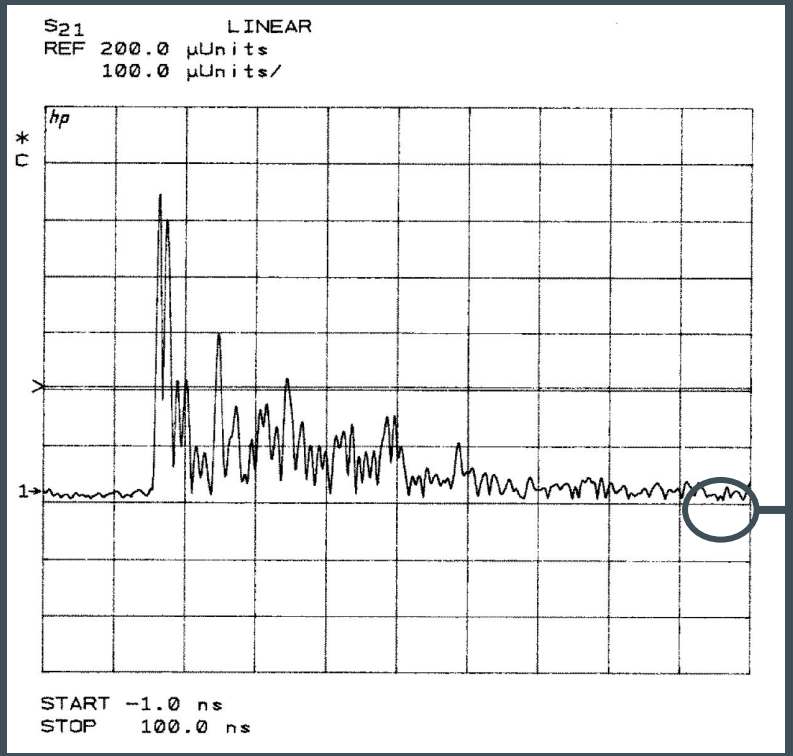
Tournament

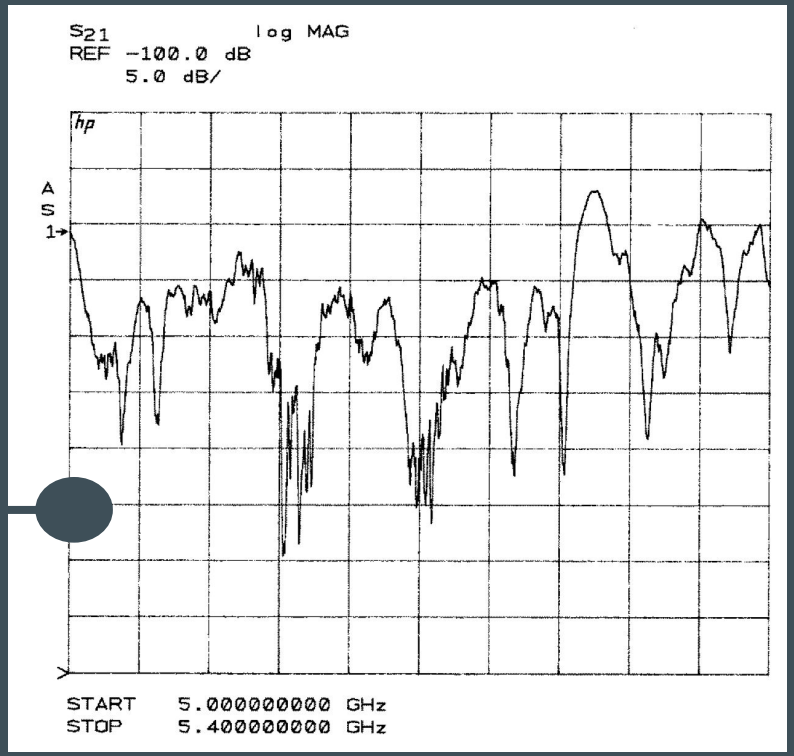
IEEE 802.11 Reference Model

- Layer 1: PHY transmission schemes
- Layer 2: MAC protocols, MAC Layer Management Entity (MLME)
- Layer 1 & 2: Station Management Entity (SME)
- Interface to higher layers: Service Access Points (SAPs)

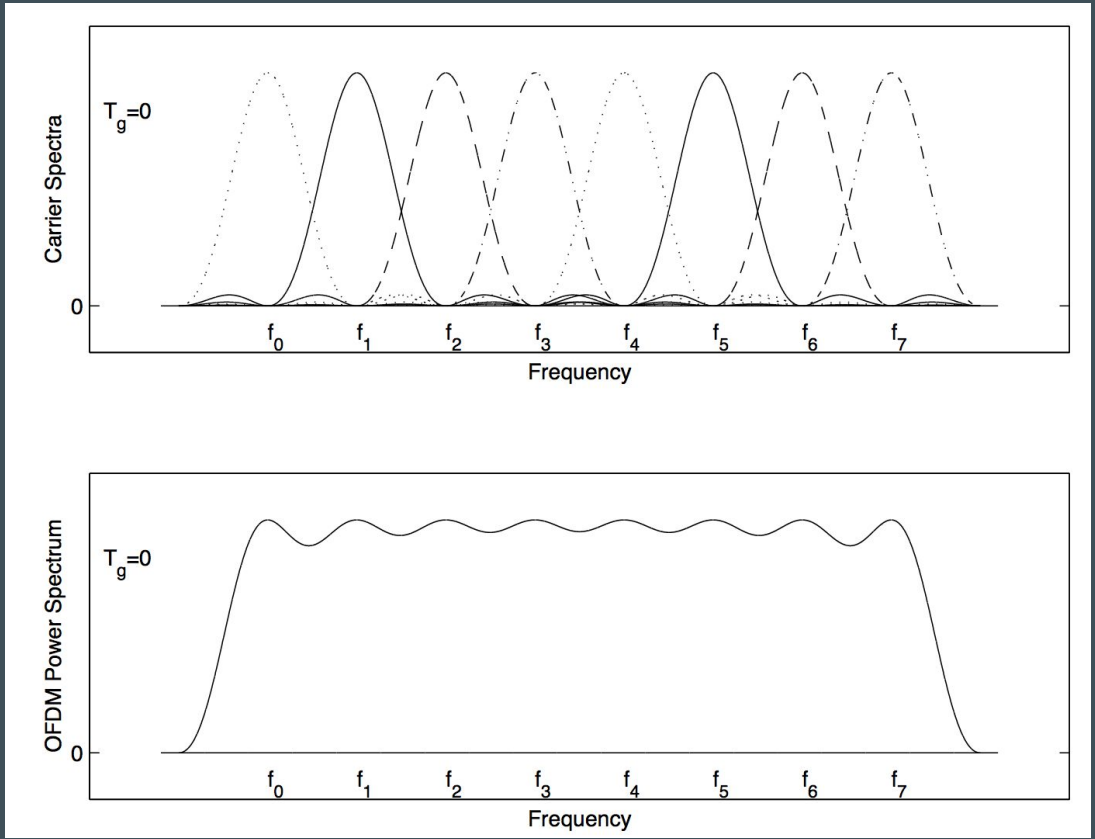


Time Domain and Frequency Domain

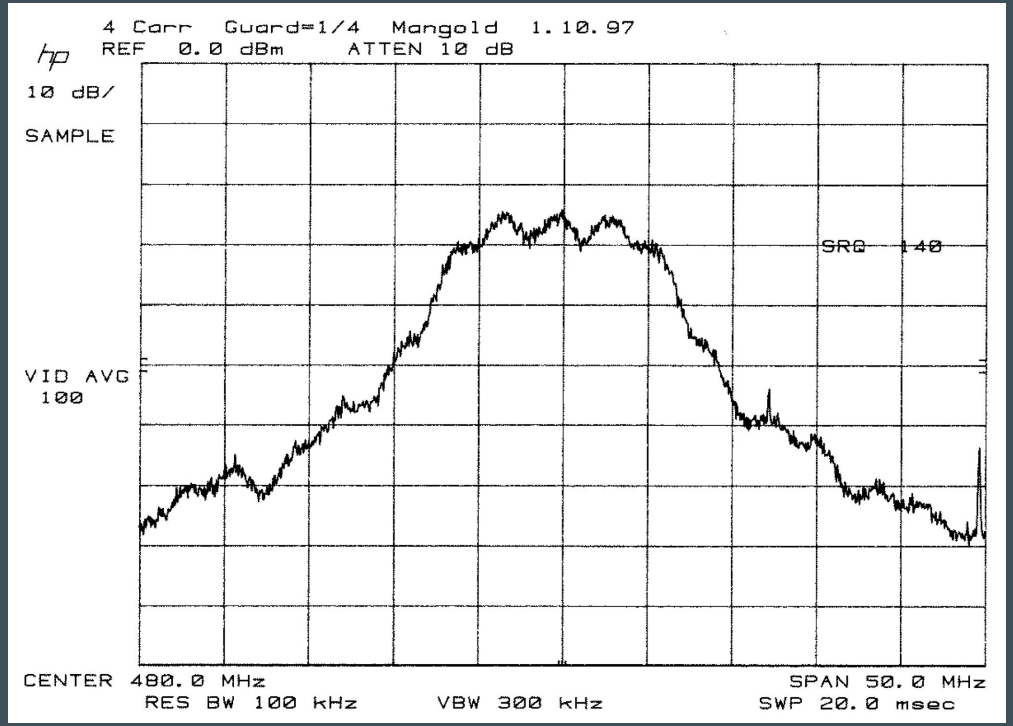




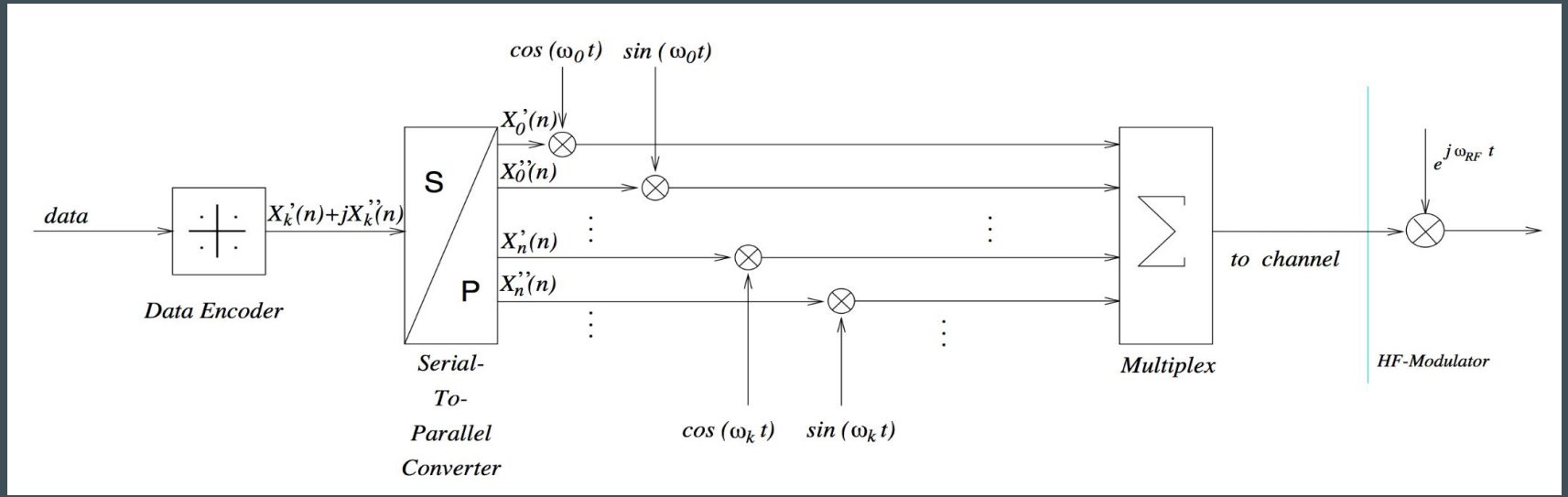
Multi-Carrier Approach



Multi-Carrier Spectrum

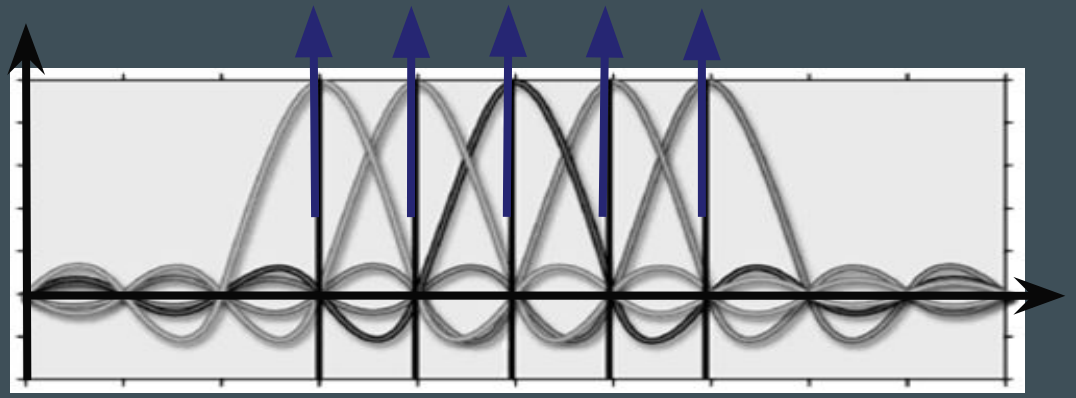


Multi-Carrier Transmitter



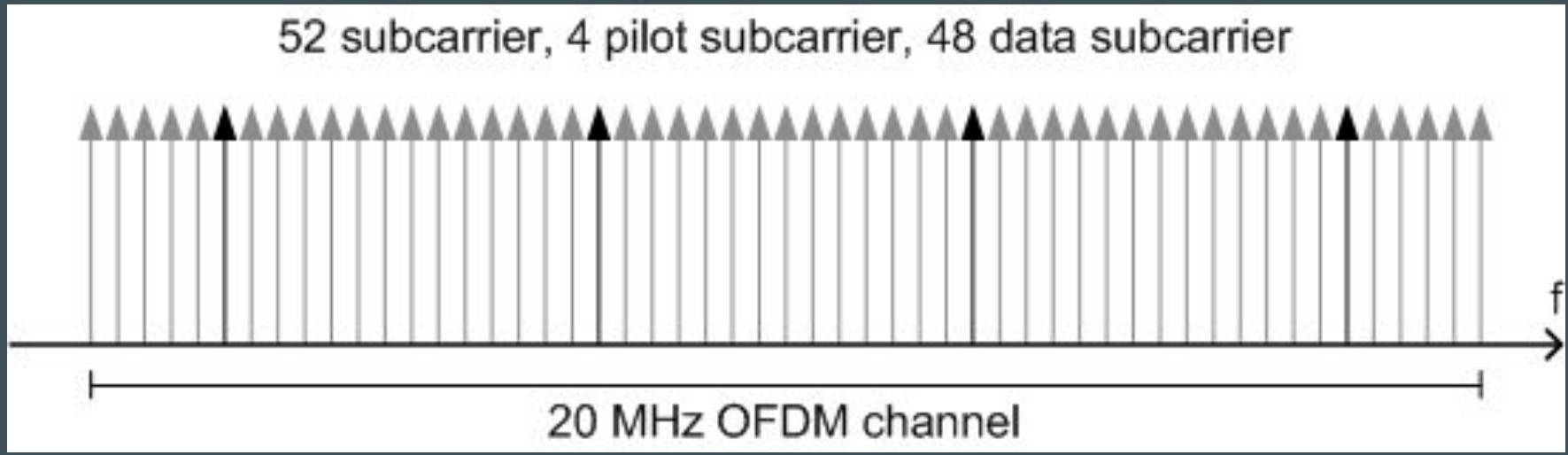
OFDM - Overview

- Orthogonal Frequency Division Multiplex
  - Bandwidth is divided into subcarriers, subcarriers overlap but remain orthogonal
- High spectral efficiency, easy implementation using (I)FFTs



OFDM in IEEE 802.11a & 11g

- Data subcarriers are used for data transmission
- Pilot subcarriers are used for synchronization



OFDM - Overview

- OFDM Subcarriers
  - Pilot subcarriers: monitoring path shifts and ICI
  - Data subcarriers: data transmission
  - Guard subcarriers: nothing transmitted
- Inter-Symbol Interference (ISI)
  - Multipath propagation: received signal arrives as an unpredictable set of reflected and direct radio waves → reason: signal delay spread
- “Inter-Channel” Subcarrier Interference (ICI)
  - Signals received from multiple indirect paths added to the direct path do not meet the condition of orthogonality between subcarriers → reason: frequency shift

Examples of OFDM Usage

- Fixed line transmission:
  - Power line communication (PLC)
  - Optical fiber communications
- Wireless:
  - IEEE 802.11a/g/n/ac/ah ... Wi-Fi Wireless LANs
  - IEEE 802.15.3a Ultra Wideband (UWB)
  - Wi-Fi 6 Wi-Fi 7
  - DAB systems, DVB terrestrial TV
- 3GPP cellular, 4G Long Term Evolution (LTE) and 5G

IEEE 802.11a/g Parameters (1/2)

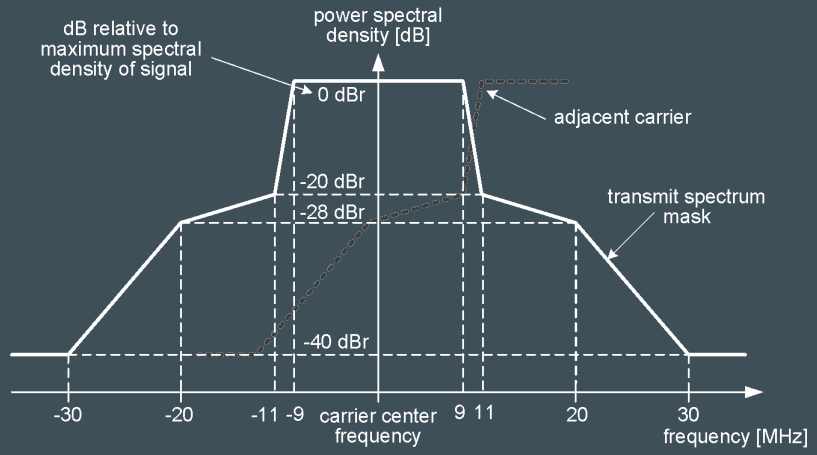
data rate [Mb/s]	modulation	coded bits per sub-carrier	coded bits per OFDM symbol	data bits per OFDM symbol
6	BPSK ½	1	48	24
9	BPSK ¾	1	48	36
12	QPSK ½	2	96	48
18	QPSK ¾	2	96	72
24	16-QAM ½	4	192	96
36	16-QAM¾	4	192	144
48	64-QAM 2/3	6	288	192
54	64-QAM¾	6	288	216

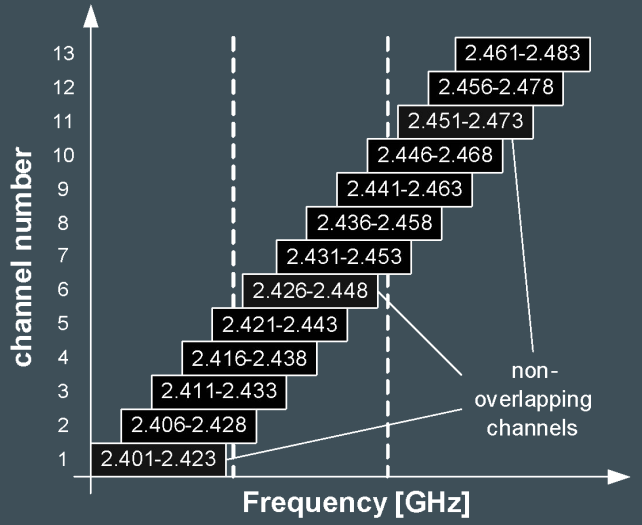
IEEE 802.11a/g Parameters (2/2)

parameter	value
Sampling rate 1/T	20 MHz
OFDM block duration Tb	64*T = 3.2 us
Guard interval duration Tg	16*T = 0.8 us
OFDM symbol duration Tb'=Tg+Tb	80*T = 4 us
Number of data sub-carriers	48
Number of pilot sub-carriers	4
Sub-carrier spacing Df	1/Tb = 0.3125 MHz
Spacing between the outmost sub-carriers	(Ntotal-1)* Df = 15.9375 MHz

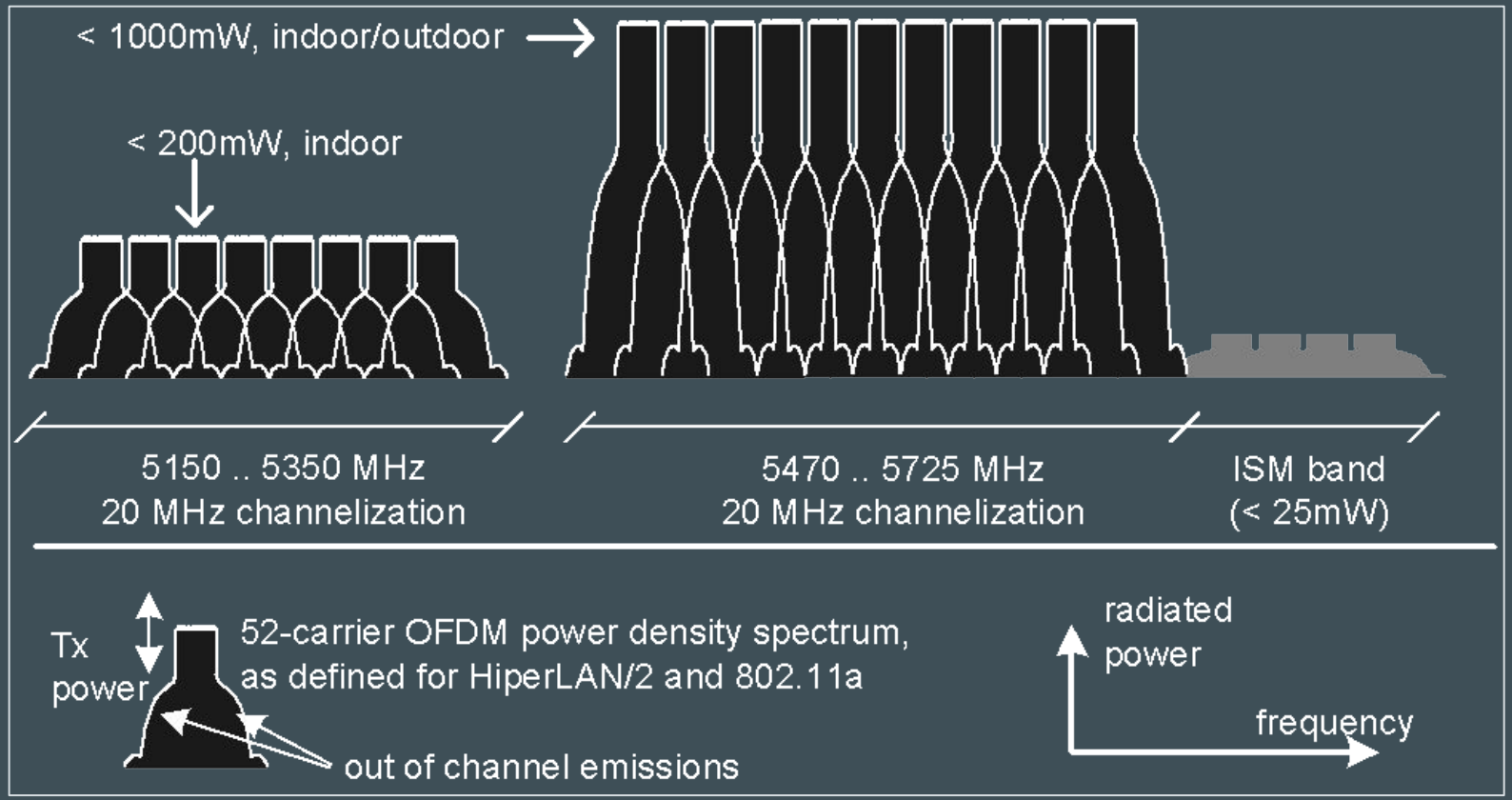
IEEE 802.11 Spectrum @ 2.4 GHz

- Transmit spectrum mask of 802.11a: limiting transmission power in frequency range
- Adjacent channels interfere with each other
  - Up to 13 channels, 3 are non-overlapping





IEEE 802.11 Spectrum @ 5 GHz



Operation @ Higher vs. Lower Frequencies

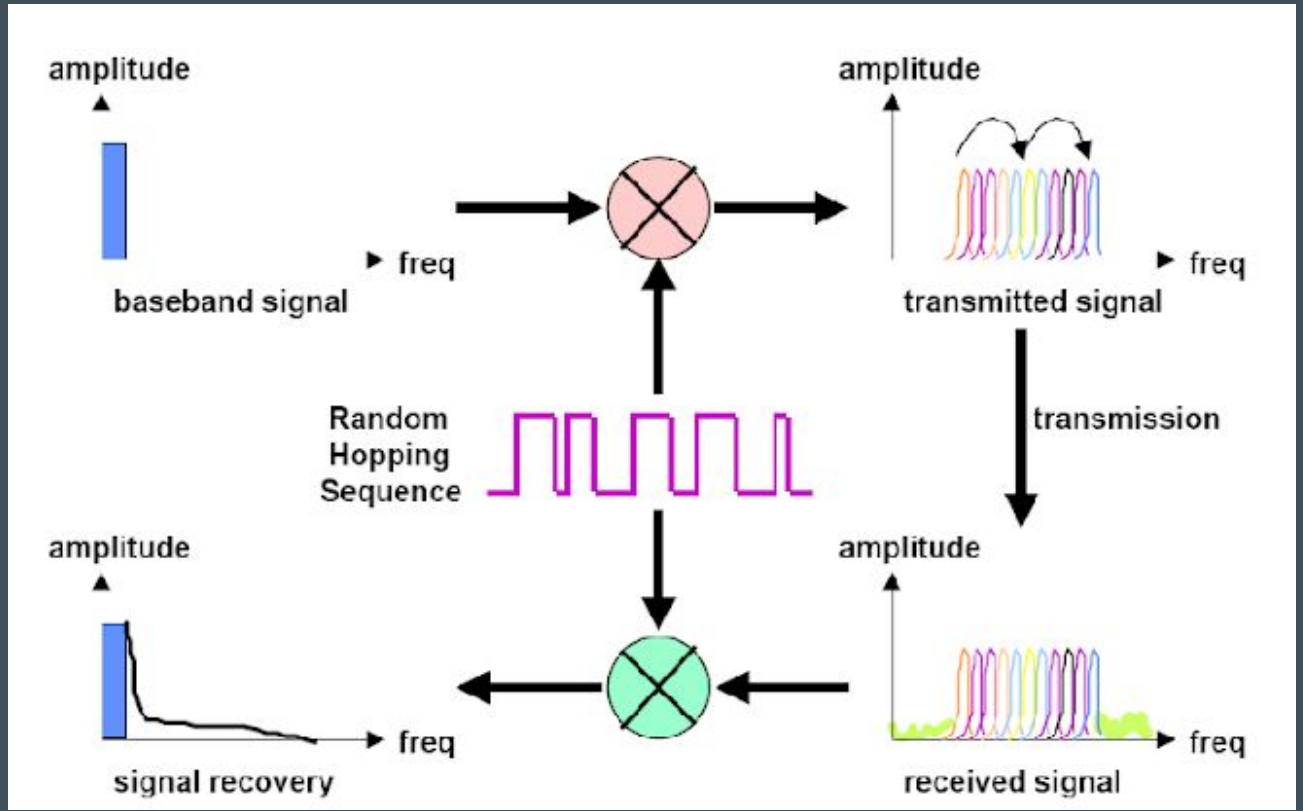
Advantages @ 2.4 GHz
- License-exempt (unlicensed) ISM band

- Disadvantages @ 2.4 GHz
  - Only 3 non-overlapping channels
- Advantages @ 5 GHz
  - 19 non-overlapping channels, higher Tx power (1W vs. 100mW)
- Disadvantages @ 5 GHz
  - Higher frequencies have higher path losses

Early / Basic Physical Layers of IEEE 802.11

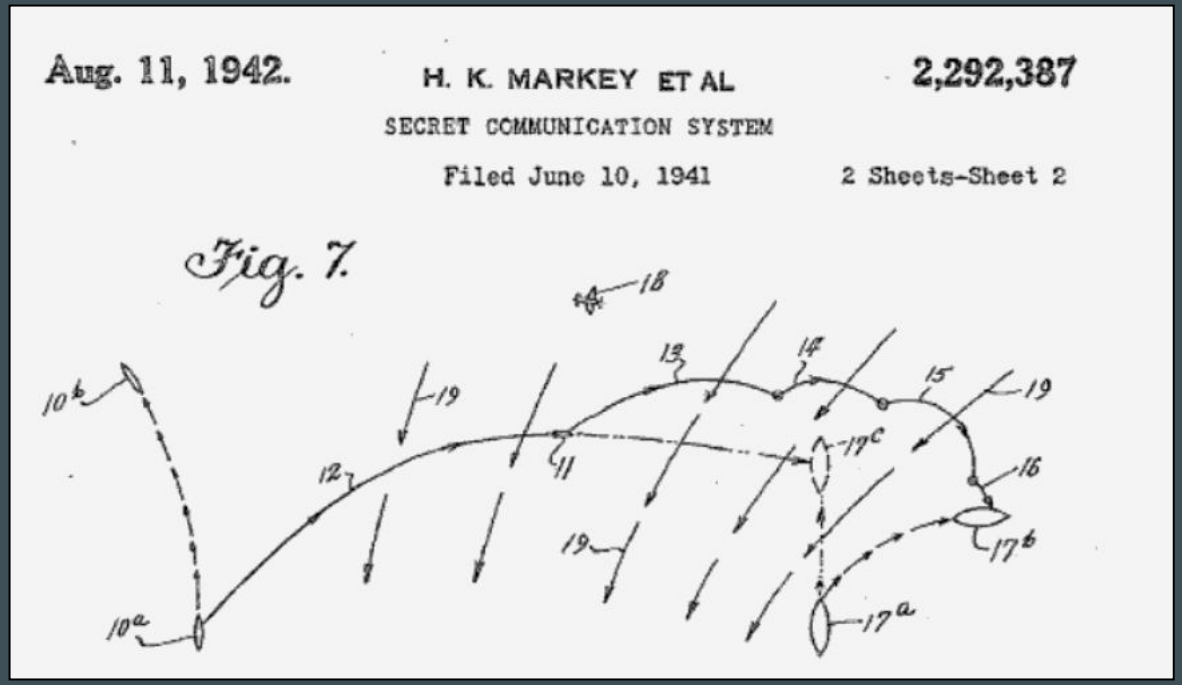
- Data rates of 1 Mb/s or 2 Mb/s
- Frequency Hopping Spread Spectrum (FHSS)
  - Frequency channel is periodically changed
  - List of selected frequencies is pseudo-random
- Direct Sequence Spread Spectrum (DSSS)
  - All stations operate at same center frequency
  - Transmitted signal is spread over full bandwidth
  - This is not CDMA: all stations operate with the same code
- Infrared (IR)

Frequency Hopping Spread Spectrum FHSS



Patent on “Secret Communication System“, 1942

Co-inventor: Hedy Lamarr, famous Hollywood actress



Aug. 11, 1942. H. K. MARKEY ET AL 2,292,387

SECRET COMMUNICATION SYSTEM

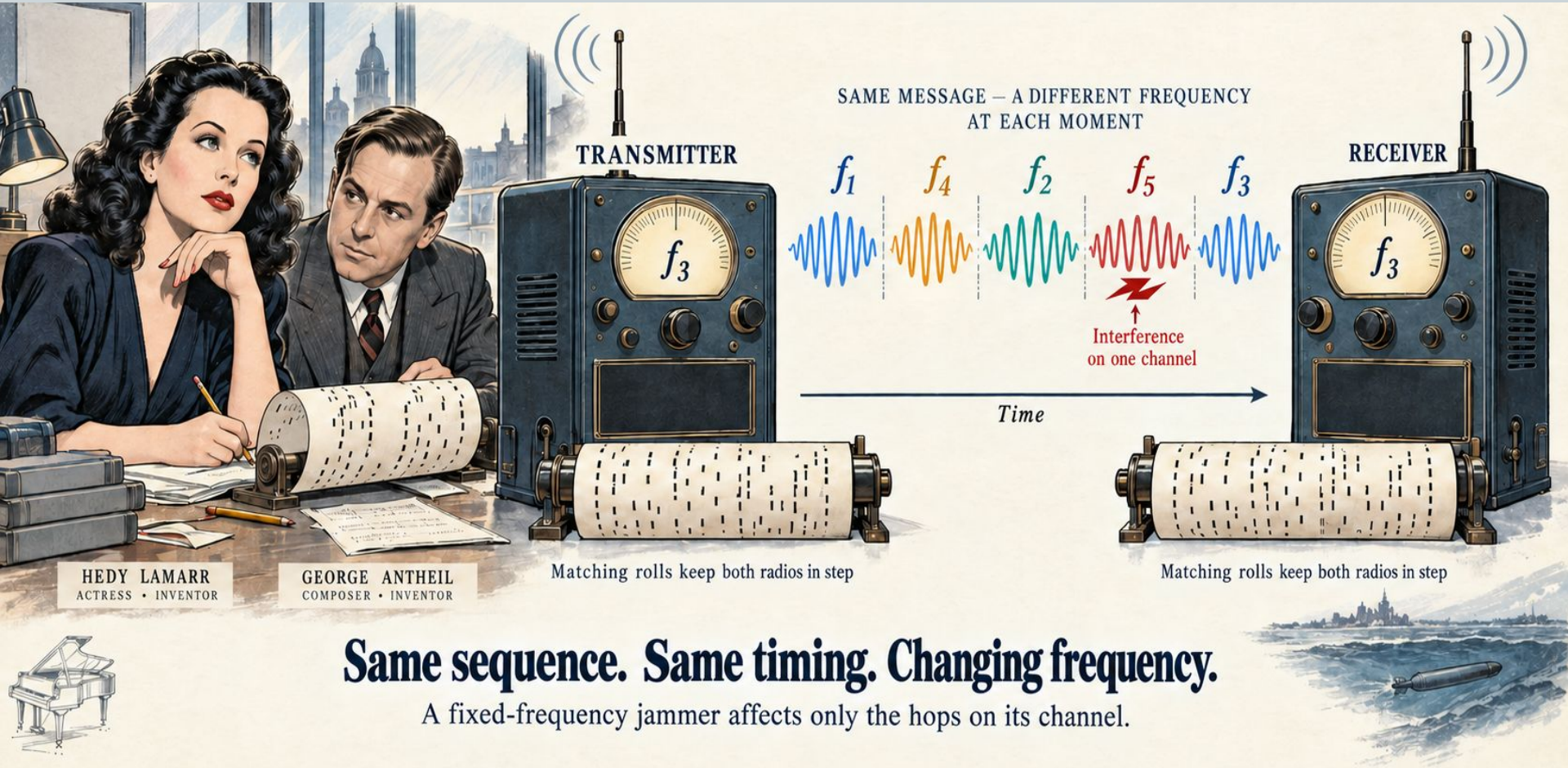
Filed June 10, 1941 2 Sheets-Sheet 2

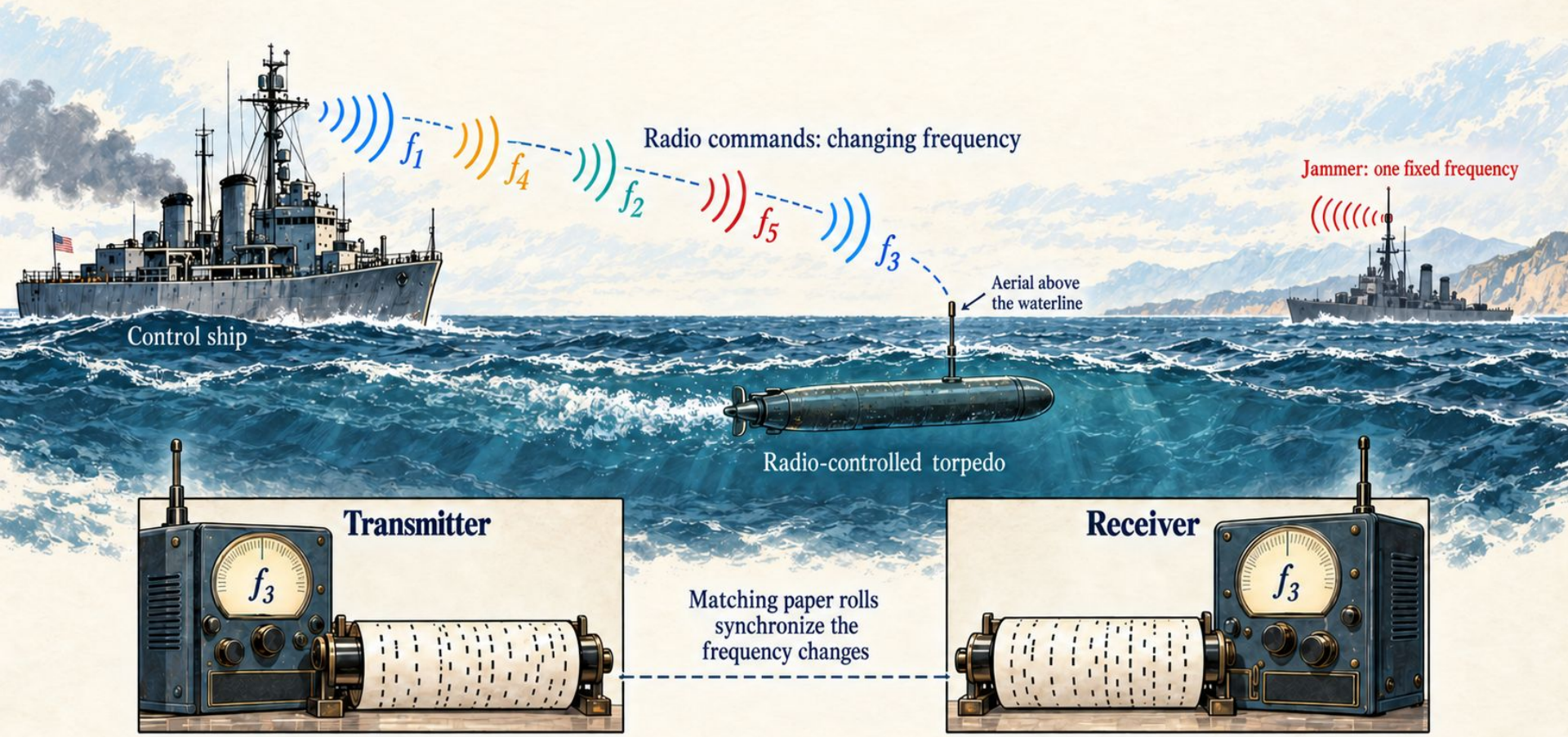
Fig. 7.





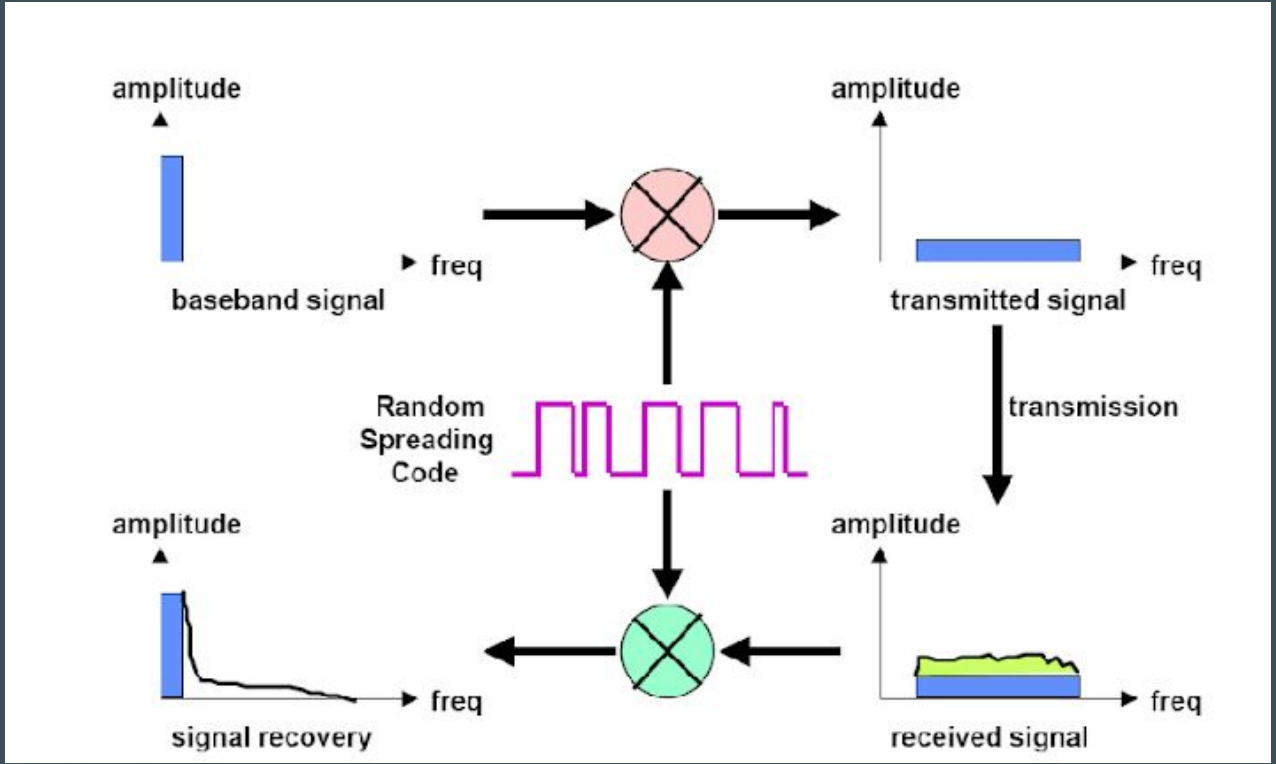
Frequency Hopping





Proposed in U.S. Patent 2,292,387; not deployed during World War II.

Direct Sequence Spread Spectrum DSSS



11 Mb/s Physical layer: IEEE 802.11b

- Extension of 802.11 @2.4GHz, in 1999
  - Raw data rates of 5.5 Mb/s or 11 Mb/s
  - Direct Sequence Spread Spectrum
  - 8-chip Complementary Code Keying (CCK) modulation
- IEEE 802.11b standard extension was the first known under the acronym Wireless Fidelity (Wi-Fi)
- The first 802.11 standard with significant commercial success

54 Mb/s Physical Layer: IEEE 802.11a

- OFDM operation @ 5GHz, 20 MHz channel bandwidth, 52 subcarriers
  - Uses 802.11 core protocol as original standard, in 1999
  - Not compatible to 802.11b

Data Rate [Mb/s]	modulation	coded bits per subcarrier	coded bits per OFDM symbol	Data bits per OFDM sysmbol	1472 byte transfer duration (µs)
6	BPSK ½	1	48	24	2012
9	BPSK ¾	1	48	36	1344
12	QPSK ½	2	96	48	1008
18	QPSK ¾	2	96	72	672
24	16-QAM ½	4	192	96	504
36	16-QAM¾	4	192	144	336
48	64-QAM 2/3	6	288	192	252
54	64-QAM¾	6	288	216	224

54Mb/s Physical Layer: IEEE 802.11g at 2.4GHz

- OFDM operation @ 2.4 GHz (same band as 802.11b)
  - Data rates: 6, 9, 12, 18, 24, 36, 48, and 54 Mb/s
  - Like 802.11a except some additional legacy overhead
- Backward compatible to non-OFDM
  - CCK for 5.5 and 11 Mb/s (802.11b)
  - DBPSK/DQPSK+DSSS for 1 and 2 Mb/s (802.11 legacy)
- Radio coverage range of 802.11g devices @2.4GHz is larger than that of 802.11a @ 5GHz

IEEE 802.11n MIMO with Multiple Antennas

- Wi-Fi 4, extension with up to 600 Mb/s
- Based on Multiple Input Multiple Output (MIMO)
- Channel bonding (= channel bundling) of two 20 MHz channels to one 40 MHz
- Operates at 2.4 GHz and 5 GHz band
- IEEE 802.11n was later extended even further towards multi-user MIMO (in IEEE 802.11ac, 2023)

Exercise: Wi-Fi Single Link Throughput

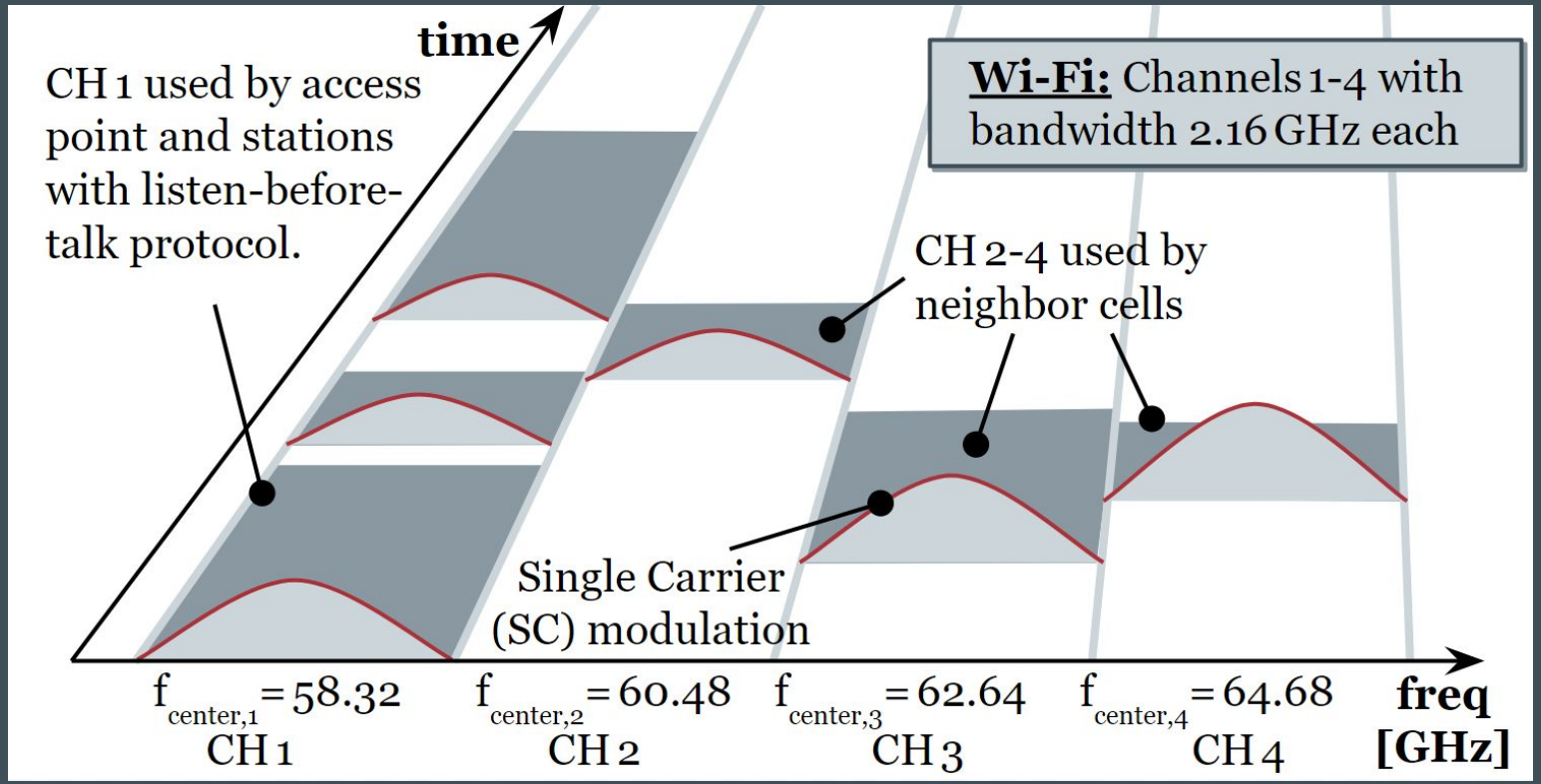
- Open in your course folder:
  02_Notebooks/03_Mobile_Systems/
  03.5_Wi-Fi_Contention_Protocol.ipynb
  03.8_Wi-Fi_Single_Link_Throughput.ipynb
- Run the cells and answer the questions.
- Save the completed notebooks and upload them to Moodle.

Summary of the Basic Modes

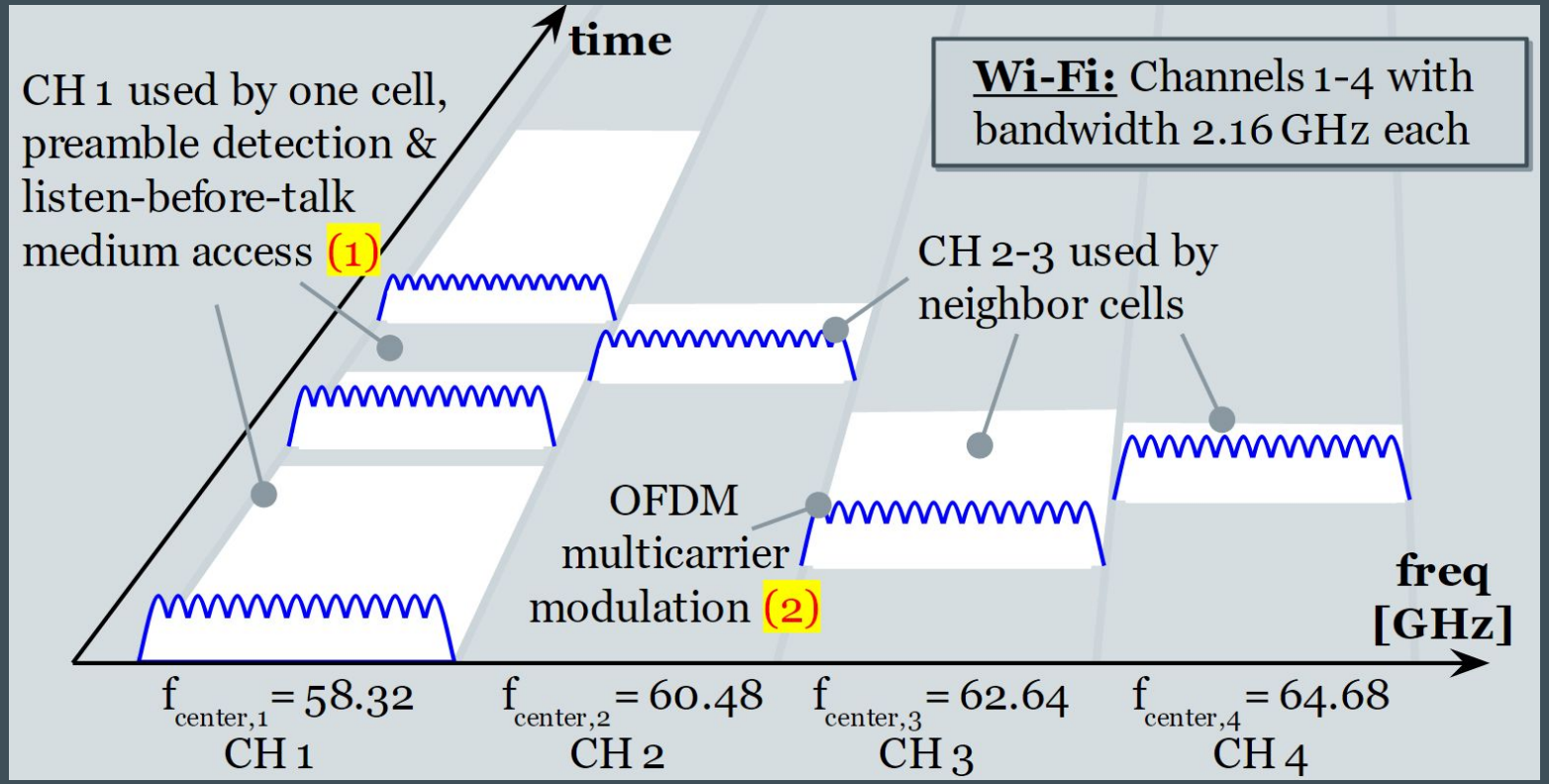
Physical Layer	Released	Frequency Band	Data Rate (max.)	Data Rate (typ.)	Range (indoor)	Range (outdoor)
Legacy	1997	2.4-2.5 GHz	2 Mb/s	0.7 Mb/s	~20 m	~75 m
802.11a	1999	5.15-5.825 GHz	54 Mb/s	23 Mb/s	~30 m	~100 m
802.11b	1999	2.4-2.5 GHz	11 Mb/s	4 Mb/s	~35 m	~110 m
802.11g	2003	2.4-2.5 GHz	54 Mb/s	19 Mb/s	~35 m	~110 m
802.11n	2007	2008.2.4 GHz and/or 5 GHz	248 Mb/s = 2x2 ant	74 Mb/s = 2x2 ant	~70 m	~160 m

- Further: 60GHz Wi-Gig, multi-user MIMO 802.11ac, TV White Space 802.11af, Internet-of-Things Wi-Fi Halow 802.11ah

Very High Throughput at 60 GHz: IEEE 802.11ad



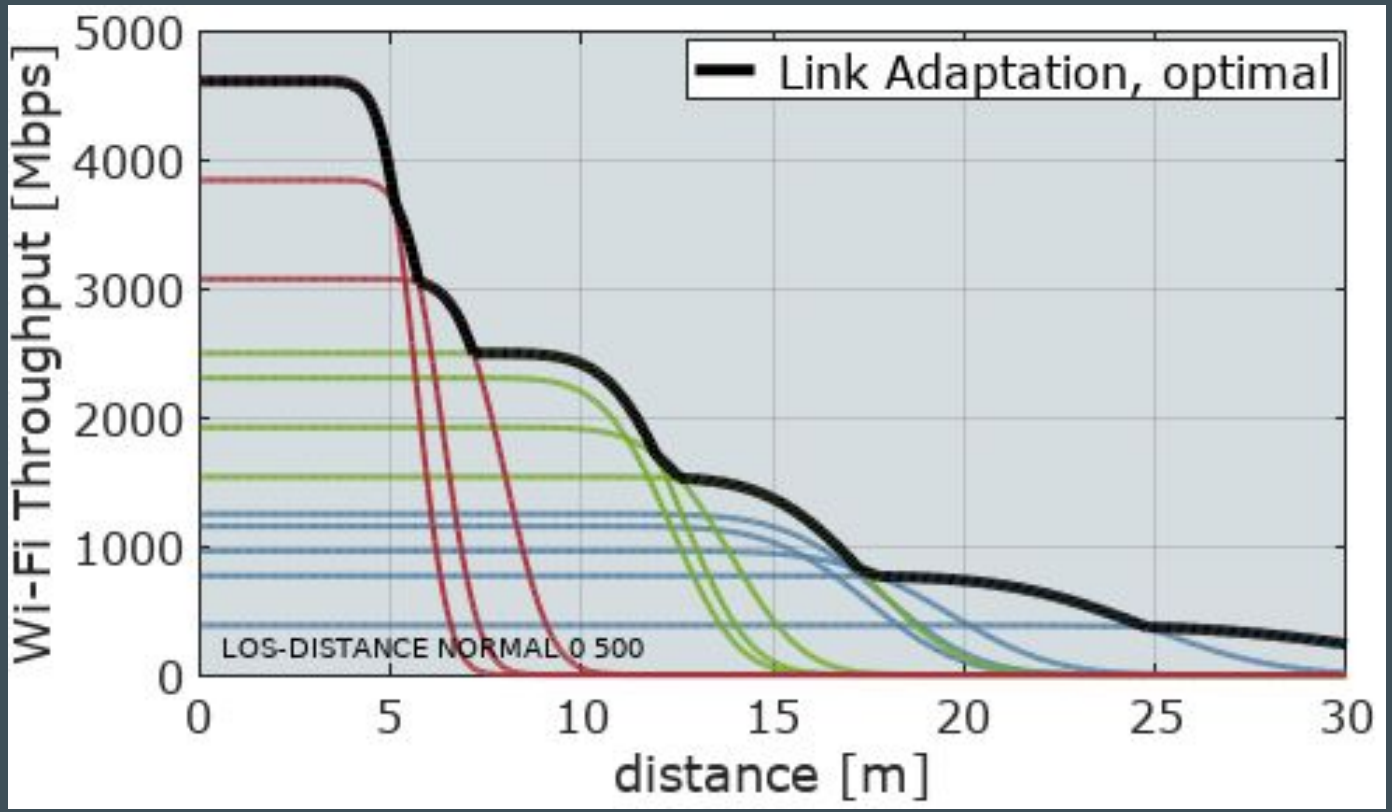
IEEE 802.11 Spectrum @ 60 GHz



IEEE 802.11ad Single Carrier Modulation - Details

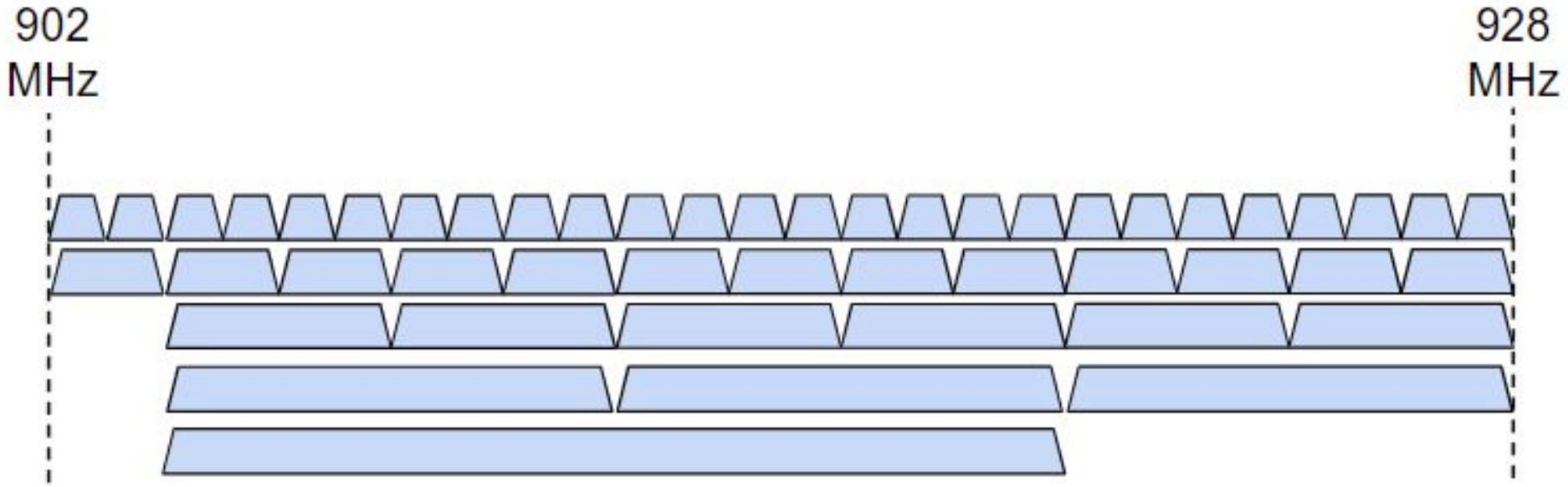
Channel bandwidth and center frequency	2160 MHz | 60.48 GHz
Signal bandwidth (-17 dBr transmit mask)	1880 MHz
Symbol rate (“SC chip rate” in 802.11ad)	1760 MHz
Symbol duration (“SC chip time” in 802.11ad)	1/1760 MHz = 0.56818 ns
Modulation	π/2-BPSK | π/2-QPSK | π/2-16QAM
Physical layer bitrate (modulated, no coding)	1760.0 Mbps | 3520.0 Mbps | 7040.0 Mbps
LDPC code rate MCS 1	1/4 (repetition, with code rate 1/2)
LDPC code rate MCS 2 ... 12	1/2, 5/8, 3/4, 13/16
LDPC code word size	672 bit
LDPC coding gain (approx.)	Rate 1/4: 8.0 dB | 1/2: 6.0 dB | 5/8: 4.0 dB Rate 3/4: 3.0 dB | 13/16: 3.2 dB
Physical layer bitrate (LDPC encoded data)	440.0 Mbps (MCS1) ... 5280.0 Mbps (MCS12)
Block size (BPSK | QPSK | 16-QAM)	448 bit | 896 bit | 1792 bit
Block duration (encoded data only)	254.55 ns
Golay sequence preceding each block	64 BPSK symbols (64 bit) | duration: 36.364 ns
Block duration (LDPC encoded data + Golay)	290.91 ns
Physical layer bitrate (LDPC encoded data + Golay)	385.0 Mbps (MCS1) ... 4620.0 Mbps (MCS12)

IEEE 802.11ad Single Carrier Performance

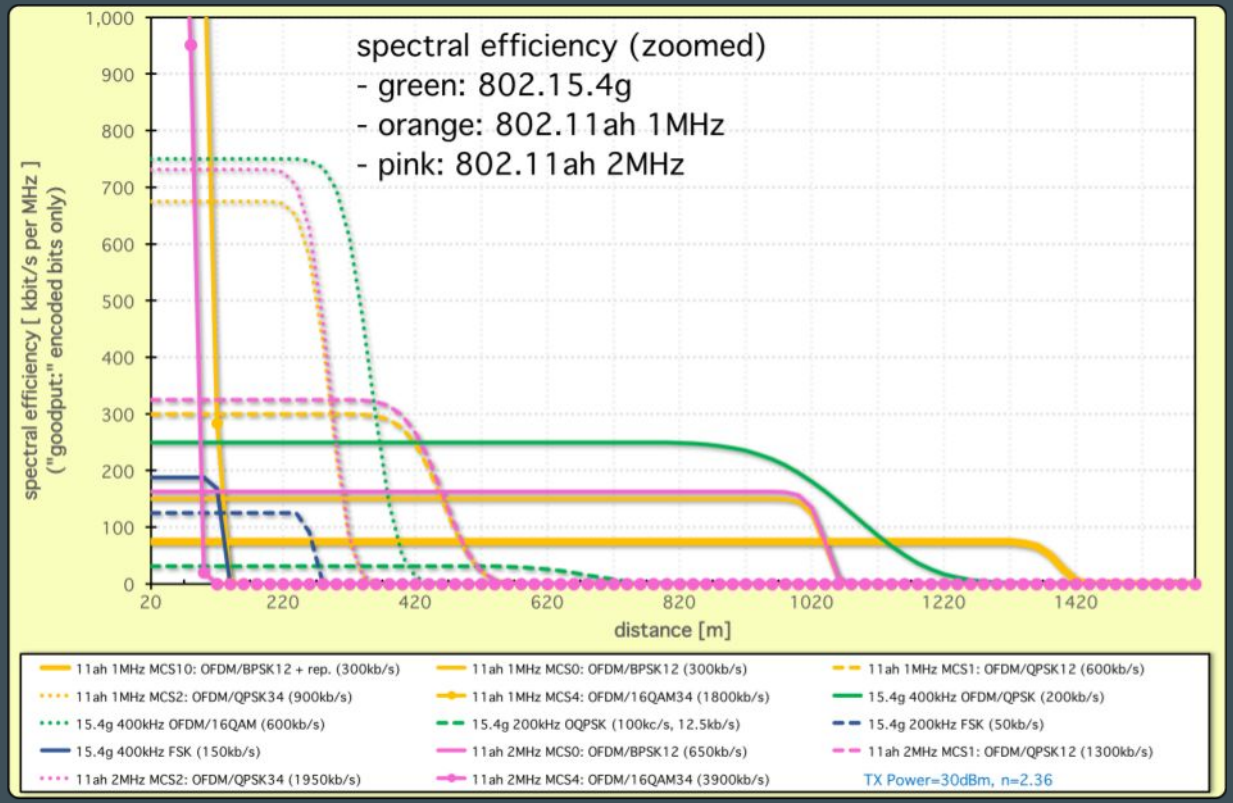


IEEE 802.11 Spectrum @ Sub 1 GHz

Wi-Fi Halow



Wi-Fi HaLow™ IEEE 802.11ah Performance



Exercise: Wi-Fi Network Performance

- Open in your course folder: 02_Notebooks/03_Mobile_Systems/ 03.6_Wi-Fi_Network_Performance.ipynb
- Run the cells and answer the questions.
- Save the completed notebook and upload it to Moodle.

Exercise: Multipath and OFDM

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.5_Multipath_and_OFDM.ipynb
- Explore multipath and run the minimal OFDM modem.
- Complete Section 5.7; vary SNR and the subcarrier count.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

